

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Power Island Interface Requirements to the EPC (utilities, chemistry, electrical, civil, transport)

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CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. Purpose

Requirements of IEC to the EPC for the design of the balance of plant around the power island. They complement the Owner's Employer's Requirements (Appendix A of ALP-EPC-001); where the Owner's requirement is stricter it prevails. Section 12 lists IEC's exceptions and clarifications to the Employer's Requirements.

2. Fuel gas at IF-01

- Pressure 36.5 barg minimum at base load, 41.0 barg maximum, rate of change ≤ 0.7 bar/s; at 36.5 barg IEC requires 45 barg at TP-G1 less the EPC station losses (EPC to confirm about 4 bar).
- Temperature 5 - 40 degC and ≥ 28 K superheat above the hydrocarbon and water dew points at all loads.
- Final filtration 3 um absolute (beta 200) with coalescing filter-separator (EPC), no free liquids.
- Trace metals: Na + K ≤ 0.02 mg/Sm3 equivalent, V ≤ 0.01 , Pb ≤ 0.02 , Ca ≤ 0.02 mg/Sm3 at IF-01; H₂S + COS ≤ 5 mg/Sm3; total sulphur ≤ 30 mg S/Sm3.
- Gas chromatograph signal (LHV, Wobbe index, H₂) to the TCS for combustion tuning (update every ≤ 4 min).

3. LDO and water injection

- LDO at IF-02 per ER-03.07 (GT inlet limits) after EPC treatment; 3.0 - 6.0 barg; 10 - 45 degC.
- Demin water for water injection at IF-03: conductivity ≤ 0.2 uS/cm, Na+K ≤ 0.02 mg/kg, silica ≤ 0.02 mg/kg; 22.5 kg/s at base load on LDO.

4. Water/steam chemistry (IAPWS TGD, VGB-S-010)

Parameter	Feedwater	HP steam	Unit
Cation conductivity (25 degC)	≤ 0.2	≤ 0.2	uS/cm
Sodium	≤ 3	≤ 2	ug/kg
Silica	≤ 10	≤ 10	ug/kg
Dissolved oxygen	≤ 20 (AVT(O))	-	ug/kg
pH (ammonia)	9.4 - 9.8	-	-
Iron / copper	≤ 5 / ≤ 2	-	ug/kg

5. Closed cooling water demand

Consumer	Duty kW	Flow m3/h	Max. supply temperature degC
GT lube oil coolers	1,600	140	38
GT generator H ₂ coolers and stator water coolers	4,300	520	38
ST lube oil coolers	850	75	38
ST generator TEWAC coolers	2,200	260	36
SFC, excitation cubicles, vacuum pump seal water	450	60	38
Total	9,400	1,055	

6. Utilities

Utility	Requirement
Instrument air	dew point -40 degC at 7 barg, 350 Nm3/h continuous (valves, purge, pulse cleaning of filter house)
Service air	150 Nm3/h intermittent
Nitrogen	HRSG and generator purge/lay-up: 3,000 Nm3 per event, 99.9 %
Hydrogen / CO ₂	H ₂ 99.9 %, 12 Nm3/day make-up; CO ₂ 150 Nm3 per purge
Auxiliary steam	4.5 t/h at 10 bar(a), 250 - 300 degC for gland sealing at start-up; HRSG sparging 3 t/h optional

7. Electrical

Consumer (IEC scope)	kW at base load, SRC
GT package auxiliaries (lube oil, hydraulic, enclosure ventilation, fuel skid, filter house)	1,180
GT generator auxiliaries (seal oil, stator water, excitation)	420
ST auxiliaries (lube oil, EHC, gland steam exhausters)	380
ST generator auxiliaries (excitation, TEWAC)	260
HRSG auxiliaries (CPH recirculation pump, ammonia skid, drains)	210

Consumer (IEC scope)	kW at base load, SRC
Condenser vacuum pump (1 running)	75
Turbine control system and UPS load	60
Total IEC auxiliaries	2,585
GT GSU losses at SRC load	1,110
ST GSU losses at SRC load	580

Largest IEC MV consumer: SFC 8.5 MW at 10.5 kV during start-up (30 min). DC: 220 V DC emergency oil pumps (GT 37 kW, ST 45 kW, 3 h autonomy). UPS 230 V AC: 45 kVA for the TCS.

8. Instrumentation and control

- IEC TCS (IE-Control T3) controls and protects GT, ST, generators and the HRSG safety functions; it is a stand-alone system connected to the plant DCS through a redundant OPC UA gateway (Modbus TCP back-up) with hardwired trips and permissives.
- Operator control of the power island from the DCS operator stations through the gateway (TCS HMI screens are also available on dedicated IEC stations in the central control room).
- Time synchronisation GPS/PTP by EPC; cybersecurity zone and conduit design per IEC 62443-3-2 by EPC with IEC input.

9. Foundations and loads

Equipment	Static load (t)	Dynamic / design data
GT + generator train (common table-top or block foundation)	1,050	unbalance and short-circuit torque (4.8 x rated), natural frequencies +/- 20 % from 50 Hz
ST + generator (spring-mounted table-top)	980	same criteria; spring elements by EPC
HRSG (steel structure base plates)	7,800 operating	wind 30 m/s, seismic PGA 0.45 g (TBDY 2018)
Condenser	1,650 operating / 2,300 flooded	side-exhaust; expansion joint to ST by IEC
GT GSU / ST GSU	395 / 245	oil pits and fire walls by EPC

10. Transport and erection

Piece	Weight (t)	Dimensions (m)	Route
GT core engine	405	13.2 x 5.9 x 5.6	barge to TP-M1, SPMT to foundation
GT generator stator	340	10.8 x 4.6 x 4.9	barge, SPMT, jacking/skidding
GT GSU	268	11.2 x 4.1 x 4.6	barge, SPMT
HRSG HP evaporator module	185	24.0 x 3.6 x 26.5	barge, SPMT, heavy crane (by EPC)
ST generator stator	185	8.6 x 4.1 x 4.2	barge, SPMT, jacking
ST GSU	168	9.4 x 3.8 x 4.3	barge, SPMT
HRSG HP drum	96	14.0 x 1.9	barge or road (under 120 t)

All pieces above 120 t exceed the road limit (ALP-OWN-GEN-001): the barge landing TP-M1 and the haul road shall take 480 t gross (SPMT) with ground bearing to be verified by the EPC.

11. Maintenance

- GT hall crane: 120 t main hook (GT rotor 112 t), 20 t auxiliary; ST hall crane: 100 t (IP/LP rotor 86 t).
- Laydown for GT major inspection: 900 m2 next to the GT, rated 10 t/m2.

12. IEC exceptions and clarifications to the Employer's Requirements

No	Requirement	IEC position
E-01	ER-05.02 evaporative cooler provision	space provision only; cooler, water supply and drain not included
E-02	SC-013 TCS 'integrated in' the plant DCS	stand-alone IEC TCS with OPC UA gateway to the EPC DCS (IEC standard, proven on the fleet)
E-03	ER-07.03 automatic ball tube cleaning	not in IEC scope (EPC with the CW system); water boxes prepared
E-04	ER-05.07 HGP >= 25,000 EOH	complies with IEC's EOH formula (starts count 20 EOH each, see DS-101 5)
E-05	Main steam, reheat and LP steam piping	by EPC (not an IEC item in SC-006 to SC-014)
E-06	H2/CO2 supply for the GT generator	by EPC
E-07	ER-04.09 house load operation 60 min	complies for the GT; ST follows on bypass operation
E-08	Guarantees	IEC guarantees are gross values of the power island (PER-001 section 7); net plant guarantees to the Owner remain with the consortium